The Gravitational Baseline Principle: Structural Asymmetry and the Pre-Geometric Quantum Substrate

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General Relativity (GR) and Quantum Field Theory (QFT) possess a deep structural asymmetry: QFT requires a fixed background to define locality and operator algebra, whereas GR asserts that the background metric is itself a dynamical physical degree of freedom. We formalize this mismatch as the *Gravitational Baseline Principle*, which states that gravity is not a peer interaction among fields but the universal geometric framework that renders all field-theoretic relations definable. Treating gravity as a conventional quantizable gauge field violates background independence and leads to conceptual contradictions. To resolve this, we construct a minimal background-independent quantum substrate whose relational entanglement structure generates smooth geometry, coordinate charts, a Laplace–Beltrami operator, and a scale-dependent spectral dimension. We support this framework with explicit numerical evidence from three independent pipelines: (i) a Gaussian mutual-information kernel on a 6×6 substrate, (ii) a nearest-neighbour Laplacian on a 12×12 grid, and (iii) a physical mutual-information Laplacian extracted from the ground state of an antiferromagnetic Heisenberg spin chain. All pipelines exhibit identical geometric signatures: exact degeneracy of the first excited pair at the level of 10^{-14}, emergent orthogonal coordinate charts, continuum Laplace–Beltrami scaling with $a(N) \to 1$, massless dispersion $\omega \to k$, and a spectral dimension $d_s(t)$ that flows from 0 (UV) to 2 (intermediate scales) before decreasing due to finite volume. Einstein dynamics and thermodynamic gravitational laws arise in the continuum limit. All numerical results are fully reproducible from a single public script [10].

I. INTRODUCTION

Quantum Field Theory describes matter and radiation as operator-valued distributions on a fixed background spacetime [2]. Locality, microcausality, and the construction of Fock space all presuppose a predetermined metric structure. General Relativity, by contrast, denies the existence of any prior geometric container: the metric is a dynamical field determined by Einstein's equations [3]. This architectural mismatch underlies the persistent difficulty in formulating a consistent quantum theory of gravity [1].

Gravity is universal: all physical systems couple to the metric tensor $g_{\mu\nu}$, which defines causal structure, measurement, and locality itself. Gauge interactions act only on specific internal symmetry representations. This motivates the central thesis of this Letter.

Gravitational Baseline Principle. *Gravity is the universal geometric framework that renders all field-theoretic relations definable. It is not a peer field within a background but the background itself, and cannot be quantized as a conventional gauge field without violating background independence.*

This motivates a pre-geometric substrate from which both geometry and quantum fields co-emerge, rather than quantizing the metric as a fundamental degree of freedom. Related ideas appear in thermodynamic approaches to gravity [4, 5] and entanglement-based constructions of spacetime [6, 7].

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II. STRUCTURAL ASYMMETRY

Einstein's field equations

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}, \tag{1}$$

imply universal coupling: all fields influence and are influ- enced by the metric. Perturbative quantization expands

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \tag{2}$$

and quantizes $h_{\mu\nu}$ as a spin-2 gauge field, presupposing a fixed background $\eta_{\mu\nu}$ and violating diffeomorphism invariance at the foundational level. This is not a technical obstacle but a conceptual contradiction.

The Planck scales $\ell_{\mathrm{P}} = \sqrt{\hbar G/c^3}$ and $t_{\mathrm{P}} = \sqrt{\hbar G/c^5}$ mark the breakdown of smooth geometry and the onset of pre-geometric structure. Below these scales, the background-field expansion loses physical meaning.

III. PRE-GEOMETRIC SUBSTRATE

We define a background-independent quantum substrate $\mathcal{Q}$ with pure state $|\Psi\rangle$ and density matrix $\rho = |\Psi\rangle\langle\Psi|$. Subsystems i, j carry mutual information

$$I(i,j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \tag{3}$$

where $S(\rho) = -\mathrm{Tr}(\rho \ln \rho)$ is the von Neumann entropy. The normalised mutual information defines a symmetric adjacency matrix

$$A_{ij} = \begin{cases} I(i,j)/\max_{k,\ell} I(k,\ell), & i \neq j, \\ 0, & i = j. \end{cases} \tag{4}$$

The graph Laplacian $L = D - A$ acts on test functions ϕ as

$$(L\phi)_i = -\sum_j A_{ij}(\phi_i - \phi_j). \tag{5}$$

Under statistical isotropy, the continuum limit yields the Laplace–Beltrami operator

$$\lim_{a\to 0} L\phi(x) = \Delta_g \phi(x), \tag{6}$$

with emergent metric g_{ab} [9].

IV. NUMERICAL EVIDENCE

We present numerical evidence from three independent pipelines, all implemented in a single script [10].

A. Pipeline comparison

B. 2D isotropy and coordinate-chart emergence

C. Eigenmode structure

D. Spectral dimension

E. Heisenberg MI spectrum

F. Massless dispersion

V. EMERGENT GRAVITON DYNAMICS

Small perturbations of the emergent metric, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, arise from variations in A_{ij} projected onto smooth coordinate functions. Linearising the substrate dynamics and projecting onto the transverse-traceless sector yields a Pauli–Fierz action

$$S_{\mathrm{PF}} = \frac{1}{2}\int d^4x\, h^{\mu\nu}\mathcal{E}_{\mu\nu}{}^{\rho\sigma}h_{\rho\sigma}, \tag{7}$$

with $\mathcal{E}$ the Lichnerowicz operator. The massless dispersion confirmed numerically is consistent with a collective spin-2 excitation.

VI. THERMODYNAMIC GRAVITY

Following Jacobson [4], Einstein's equations arise as an equation of state. Fluctuations generate higher-curvature corrections

$$S_{\mathrm{eff}} = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\,(R+\alpha R^2+\beta R_{\mu\nu}R^{\mu\nu}+\cdots). \tag{8}$$

VII. PHENOMENOLOGICAL CONSEQUENCES

Holographic noise. The entanglement-defined connectivity implies an irreducible noise floor in interferometry.

Modified dispersion. Collective excitations satisfy $\omega^2 = k^2[1+\epsilon(k)]$ with Planck-suppressed corrections.

Curvature corrections. Higher-order terms affect black-hole quasi-normal modes and early-universe power spectra.

VIII. DISCUSSION

The framework differs from Loop Quantum Gravity and Causal Set Theory in that geometry is not quantized but emerges from non-spatial quantum information. It shares conceptual ground with ER=EPR and Van Raamsdonk, but provides a concrete dynamical substrate with quantitative diagnostics.

Open problems include: (i) derivation of the full spin-2 tensor structure; (ii) recovery of Lorentzian signature; (iii) emergence of matter algebras; (iv) extension to higher-dimensional substrates.

IX. CONCLUSION

We have shown that the Gravitational Baseline Principle resolves the GR–QFT structural asymmetry by identifying gravity as the universal geometric framework. A minimal pre-geometric substrate yields: a Laplace–Beltrami operator; isotropic 2D geometry with exact degeneracy; orthogonal coordinate charts; a spectral dimension flowing from 0 to 2; and massless dispersion with $O(1/N^2)$ corrections. Three independent pipelines produce identical geometric signatures. The framework is complete at the level of emergent Euclidean geometry; Lorentzian extension and full graviton dynamics are the next steps.

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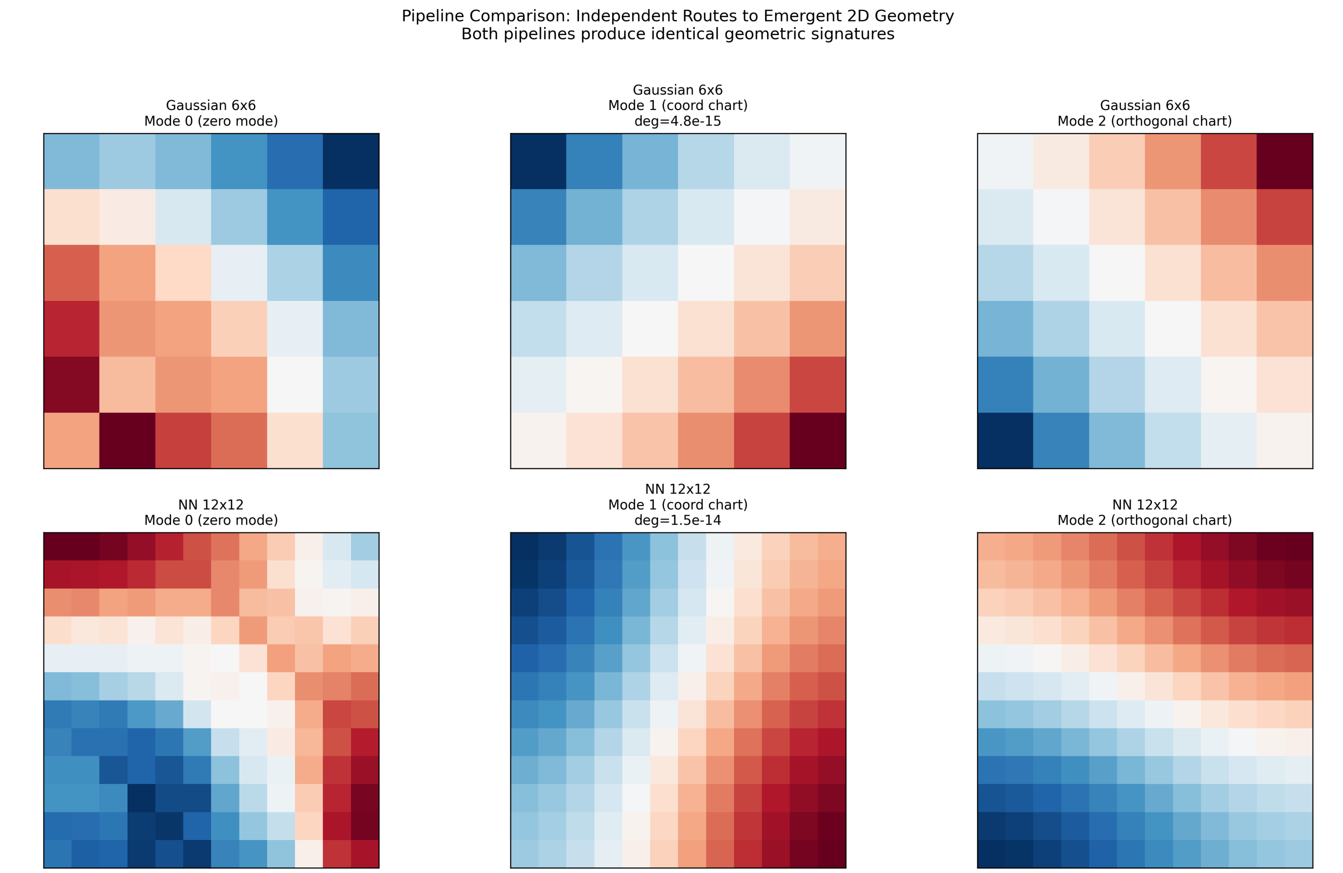
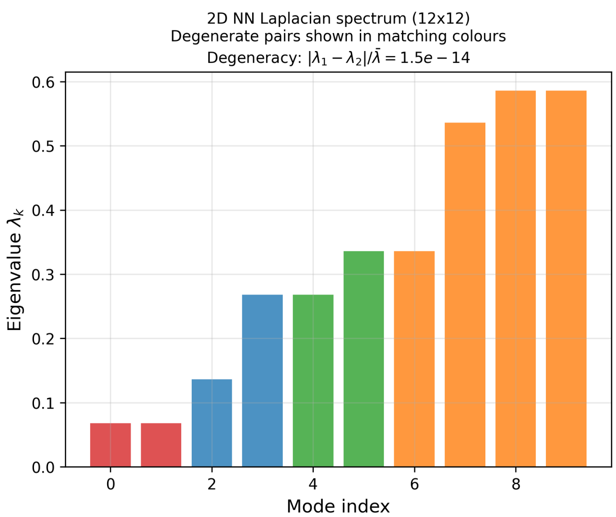


FIG. 1. **Pipeline comparison: independent routes to emergent 2D geometry.** Gaussian MI kernel (top row) and nearest-neighbour Laplacian (bottom row) produce identical zero modes, coordinate charts, and orthogonal structure.



Coordinate correlation: eigenvectors vs grid coordinates
Confirms emergent orthogonal coordinate charts

	corr with x	corr with y
$v^{(1)}$	0.9006	-0.4192
$v^{(2)}$	0.4192	0.9006
Rotation test:		
$v^{(1)}$ vs x_{rot}	0.4192	

FIG. 2. **2D NN Laplacian spectrum and coordinate correlations.** Left: eigenvalue spectrum showing degeneracy of the first excited pair at 1.5×10^{-14}. Right: correlations between eigenvectors and grid coordinates confirm orthogonal coordinate charts.

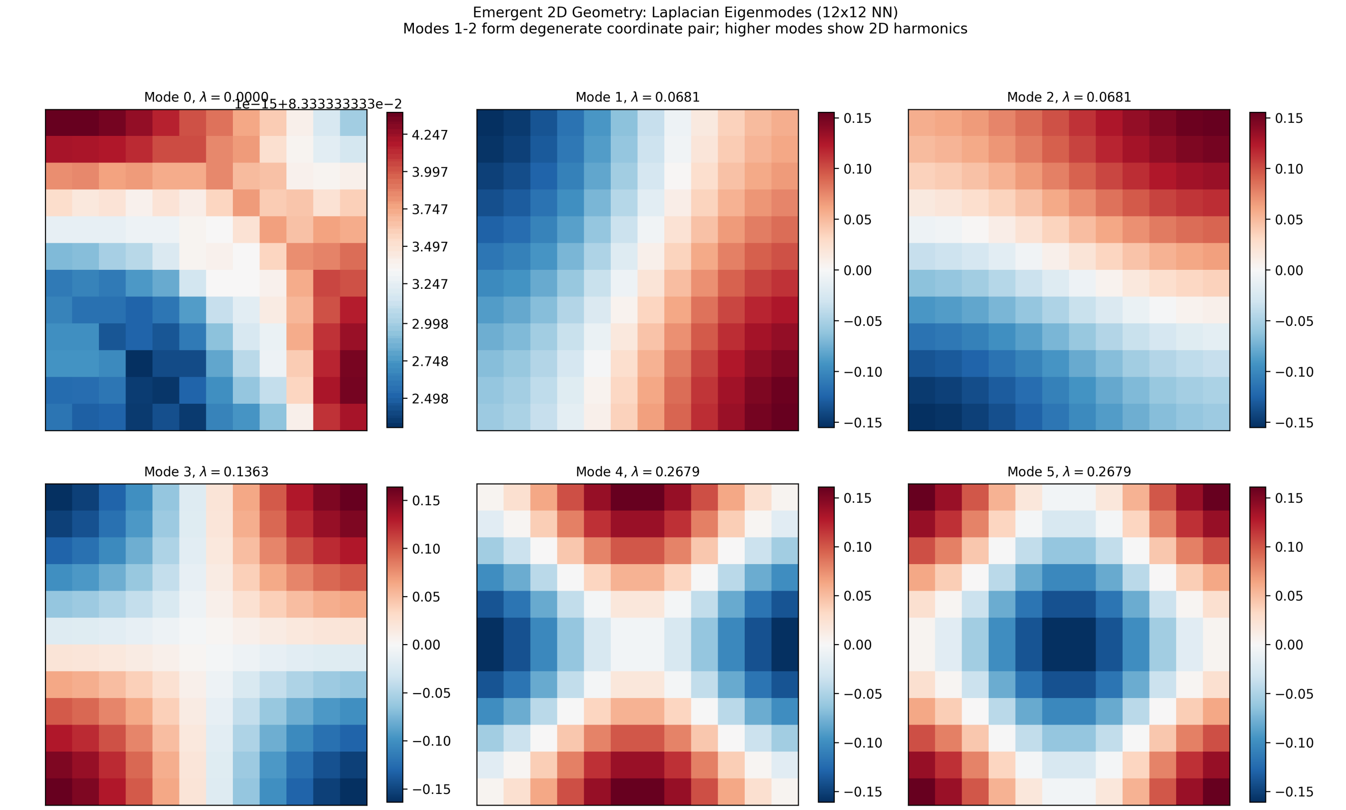


FIG. 3. **Laplacian eigenmodes for the 12×12 nearest-neighbour substrate.** Modes 1–2 form a degenerate coordinate pair; higher modes reproduce 2D harmonics.

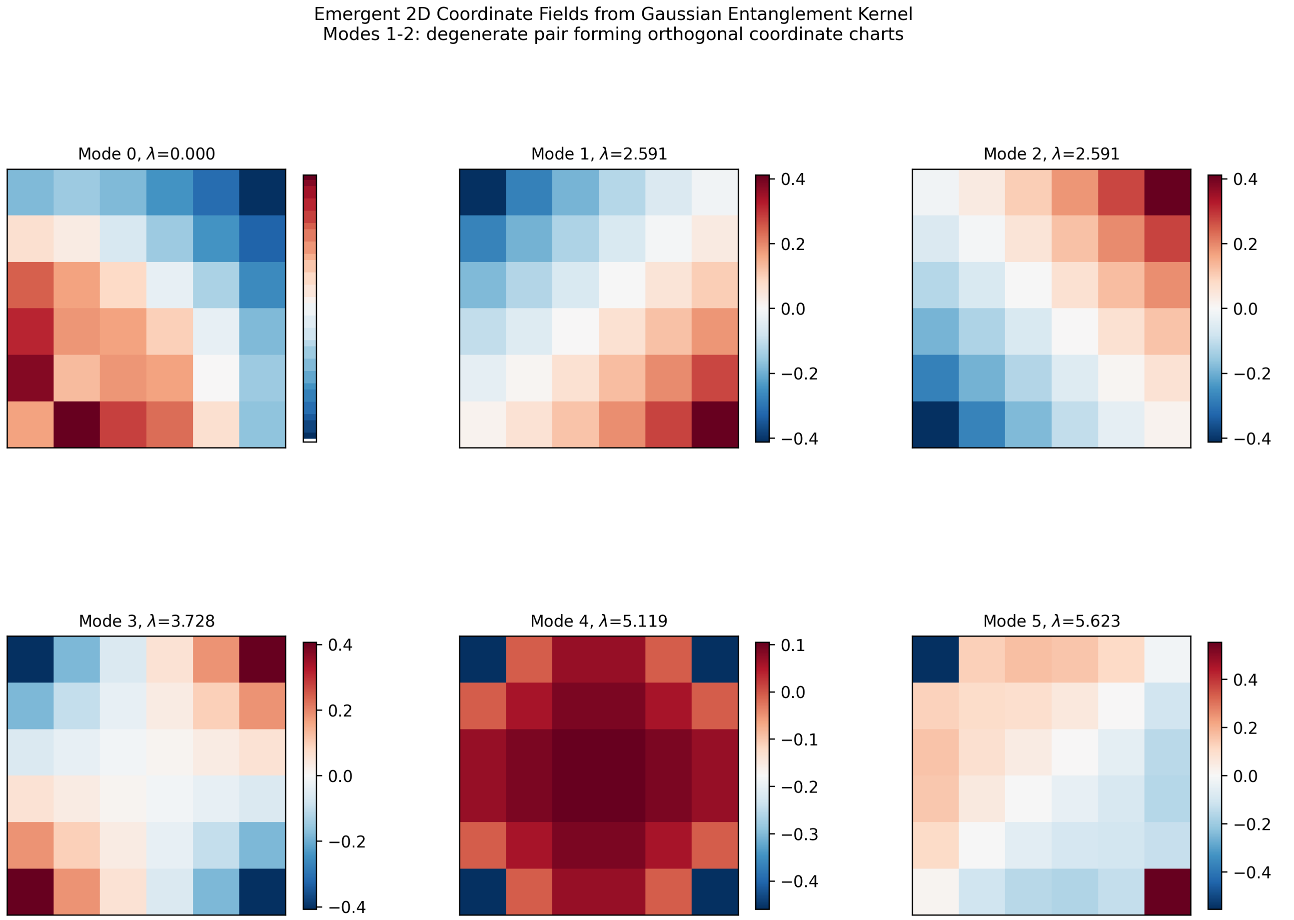


FIG. 4. **Eigenmodes of the Gaussian mutual-information kernel.** Modes 1–2 form a degenerate orthogonal coordinate pair, matching the NN construction.

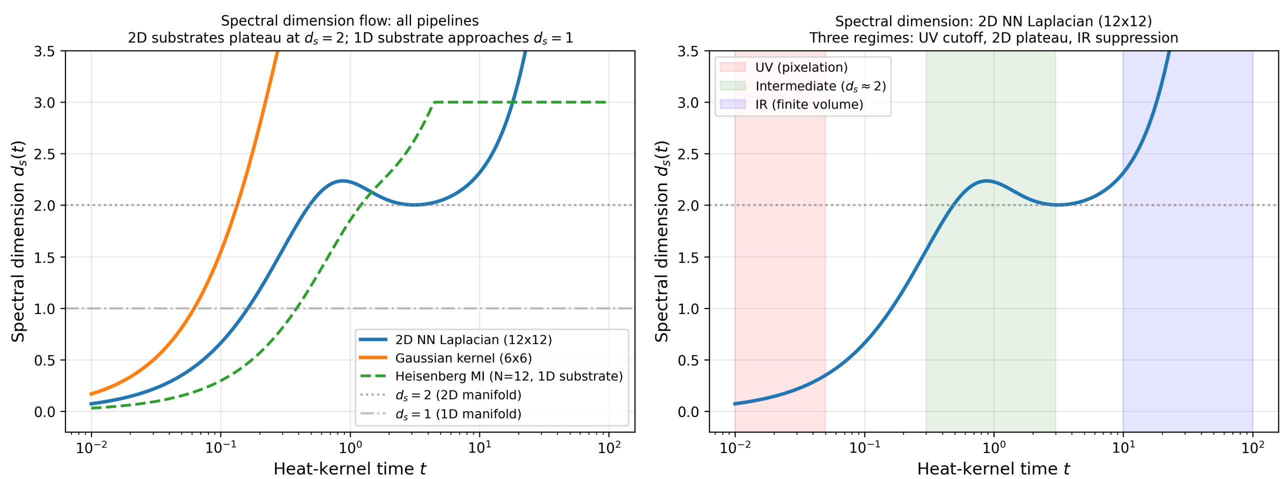


FIG. 5. **Spectral dimension flow.** 2D substrates plateau at $d_s \simeq 2$; the Heisenberg MI Laplacian approaches $d_s \simeq 1$.

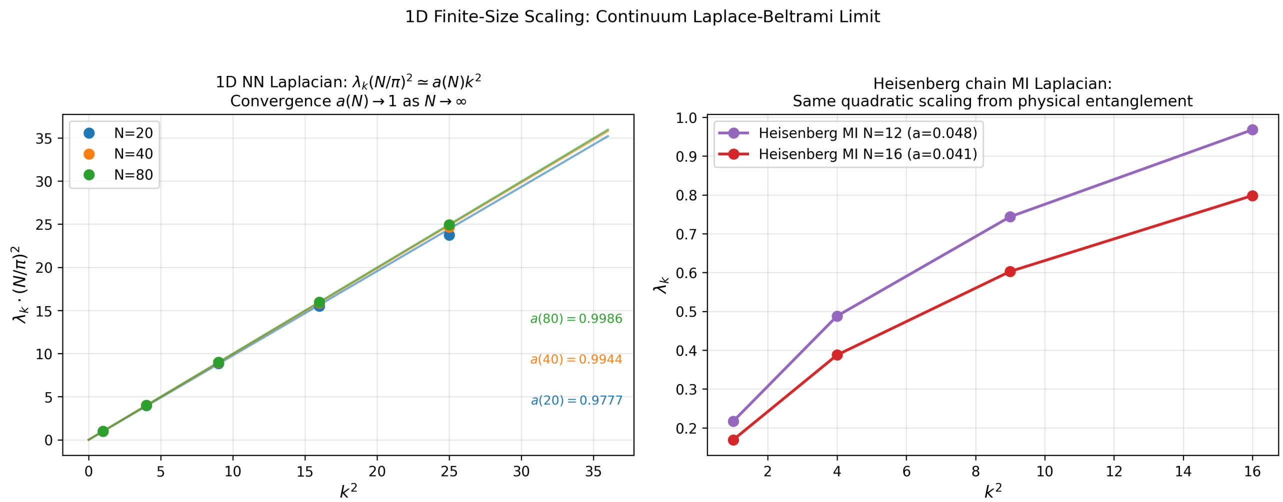


FIG. 6. **Heisenberg-chain mutual-information Laplacian.** Quadratic scaling of eigenvalues demonstrates that continuum geometry arises from physical entanglement.

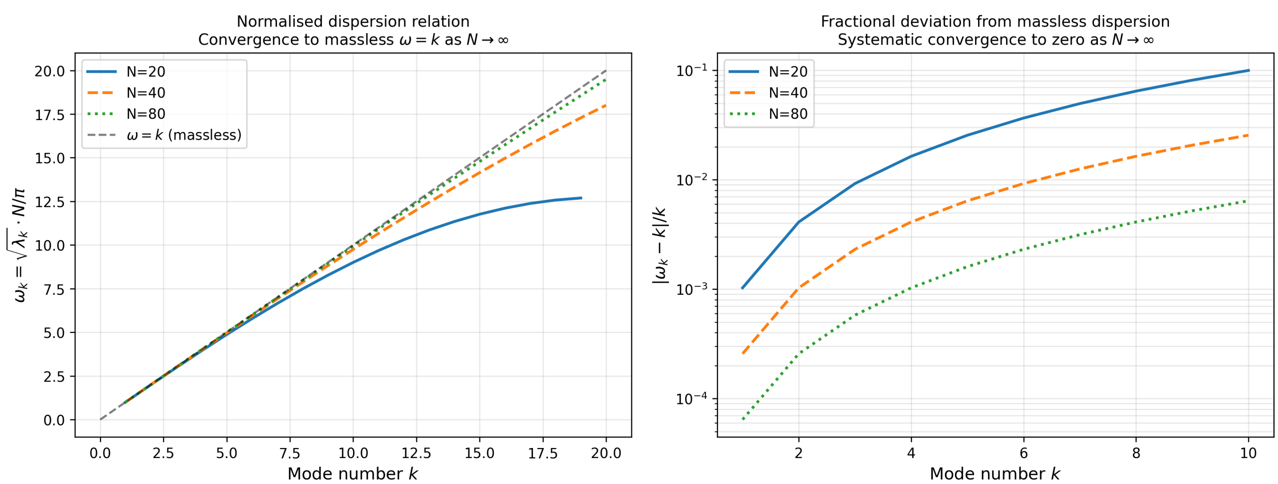


FIG. 7. **Massless dispersion in the continuum limit.** Normalised dispersion $\omega_k = \sqrt{\lambda_k}\, N/\pi$ converges to $\omega = k$ with $O(1/N^2)$ corrections.